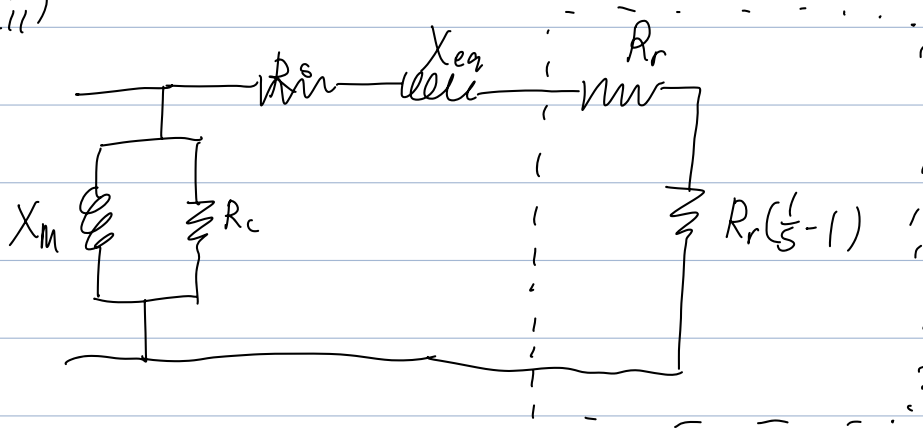


Q3.

$$(i) N_s = \frac{60 f}{P/2} = 1500 \text{ rpm}$$

$$\text{Slip } s = \frac{N_s - N_r}{N_s} = \frac{1}{30} \approx 0.033$$

(ii)



$$Z_m = R_s + \frac{R_r}{s} + X_{eq} j = 15 + \boxed{30 \times 2} + 6j = 61.79 \angle 5.57^\circ \Omega$$

$$I_s = \frac{V_{in}}{Z_m} = \frac{230}{61.79} = 3.72 \angle -5.57^\circ \text{ A} \quad 62.4 \angle 5.52^\circ \Omega$$

$$P_{\text{air-gap}} = I_s^2 \cdot \left(\frac{R_r}{s}\right) = \boxed{825.22} \times 3 = 2475.65 \text{ W}$$

$$(iii) P_{\text{mech}} = (1-s) P_{\text{air-gap}} = 802.6 \text{ W} \times 3$$

$$(iv) T = \frac{P_{\text{mech}}}{\omega} = \frac{802.6}{1450 \times \frac{2\pi}{60}} = 5.29 \text{ Nm} \times 3$$

$$(v) P_{\text{loss}} = \cancel{I_s \cdot V_{in}} - P_{\text{mech}} = 53 \text{ W} \times 3$$

$$(vi) \eta = \frac{P_{\text{mech}}}{\cancel{I_s V_{in}}} = 93.81\%$$

$$(vii) \cancel{P_{in}} = \cancel{I_s V_{in}} = 855.6 \text{ W} \times 3$$

$$(viii) I_s = 3.72 \angle -0.97^\circ \text{ A}$$

PF :